

LAB WORKSHEET

Lab 21 — Multi-Factor Authentication with TOTP (PAM)

Name		Date	
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Objective: 3.3 — Multifactor authentication: something you know / have / are.

Work through the lab on the Killercoda Ubuntu playground and record your results below.

Task 1 — Provision

Create mfauser and generate the TOTP secret.

Paste the first 3 lines of `.google_authenticator`.

Task 2 — PAM wiring

Wire PAM and sshd for TOTP.

Record the `AuthenticationMethods` line used.

Task 3 — Generate code

Generate the current code with `oathtool`.

Record a code (note it expires in ~30 s).

Task 4 — Test login

Attempt an SSH login with a wrong code, then a right one.

Record that the wrong code denied access.

Task 5 — Factor test

Explain why password + scratch code is NOT MFA.

State the factor-category rule.

Completion checklist

- Provisioned a TOTP secret for a user.
- Wired PAM and sshd to require two factors.
- Generated a TOTP code from the CLI.
- Confirmed a wrong code denies access.
- Applied the factor-category test for real MFA.